Virtual Experiment 1

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PH24C031

1 Introduction

In the study of nuclear radiation, understanding the half-lives of radioactive isotopes is crucial for various applications, ranging from medical treatments to nuclear power generation. The half-life of a radioactive substance is the time required for half of the atoms in a sample to decay. This concept is fundamental to predicting the behavior of radioactive materials over time and is essential for both practical applications and theoretical research.

In this experiment, we focus on determining the half-lives of various radioactive sources commonly used in laboratory settings: Americium-241 (^{241}Am), Strontium-90 (^{90}Sr), Barium-133 (^{133}Ba), and Protactinium-234 (^{234}Pa). These isotopes emit different types of radiation—alpha, beta, and gamma—which are detected using a Geiger-Müller counter.

Protactinium-234, with its relatively short half-life, provides an opportunity to observe radioactive decay in real-time within a controlled laboratory environment. The other sources—^{241}Am, ^{90}Sr, and ^{133}Ba—each have distinct half-lives and emit different types of radiation, making them useful for a variety of experiments.

The determination of the half-life involves fixing the operating voltage, the distance of the detector from the source, and the counting interval, followed by careful observation and measurement of the decay process. This experiment not only helps in understanding the decay behavior of different isotopes but also in appreciating the importance of selecting the right source for specific experimental needs.

The following sections will detail the procedures for estimating the half-life of Protactinium-234 and discuss the half-lives of ^{241}Am, ^{90}Sr, and ^{133}Ba, providing explanations for their suitability in various experimental settings.

2 Working of GM Counter

The Geiger counter would contain a Geiger-Müller tube, the element of sense that detects the radiation and the electronics that process and would provide the result.

The Geiger-Müller tube is filled with a gas such as helium, neon, or argon at the pressure being the lowest, where there is an application of high voltage. There would be the conduction of the electrical charge on the tube when a particle or photon of incident radiation would turn the gas conductive by the means of ionization.

2.0.1 Types of Geiger Counter

The Geiger counter is dictated entirely by the design of the tube and can be generally categorised into two types:

- **End Window**
- **Windowless**

2.0.2 End Window

This style of the tube would have a small window at one of its ends. This window would be helpful in ionizing particles that could travel easily.

2.0.3 Windowless

As the name suggests, this type of tube would not have any windows and the thickness would be in the range of one to two mm. This type of tube is used for detecting high penetrating radiations.

2.0.4 Geiger Counter Units

The measurement of particles would be in different units, the widely used one of them is the Counts Per Minute (CPM). The measurement of radioactivity would be in micro-Sieverts per hour (µSv/hr) and milli-Roentgens per hour (mR/hr).

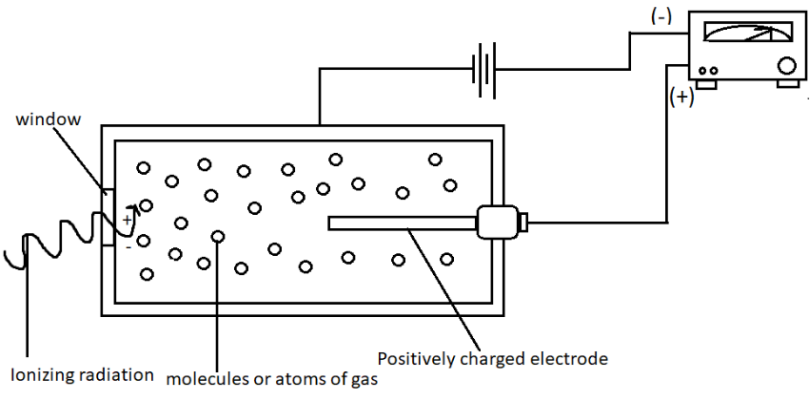


Figure 1: Circuit Diagram

2.0.5 Operating Voltage

The operating voltage of a Geiger-Müller (GM) counter typically ranges between 400 to 1200 volts, depending on the specific design and the type of gas used inside the tube. The operating voltage is the voltage at which the GM tube detects ionizing radiation and generates a detectable pulse.

Below the operating voltage, the tube does not detect radiation efficiently, and very few pulses are generated. Above the operating voltage, the tube may produce continuous discharges, leading to inaccurate readings or even damage to the tube. The exact operating voltage is determined experimentally by finding the "plateau" region on a graph of count rate versus voltage, where the count rate remains relatively stable over a range of voltages.

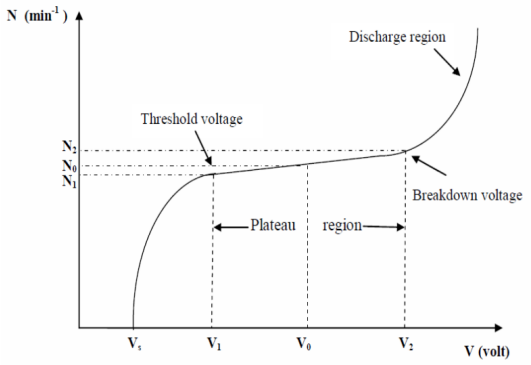


Figure 2: Count vs Voltage plot

2.1 Operating Voltage data:

All datas are taken from Radiation Lab Virtual Environment.

2.1.1 For Alpha Particle

Shield Thickness (mm)	Distance (cm)	Count Duration (sec)		Voltage	Count
0	6	10		500	10
0	6	10		525	47
0	6	10		550	173
0	6	10		575	561
0	6	10		600	1738
0	6	10		625	4858
0	6	10		650	11790
0	6	10		675	22516
0	6	10		700	36348
0	6	10		725	50080
0	6	10		750	60597
0	6	10		775	67398
0	6	10		800	71383
0	6	10		825	72160
0	6	10		850	72790
0	6	10		875	72741
0	6	10		900	72820
0	6	10		925	73283
0	6	10		950	73461
0	6	10		975	73360
0	6	10		1000	73917
0	6	10		1025	74283
0	6	10		1050	74607
0	6	10		1075	74471
0	6	10		1100	75000
0	6	10		1125	75123
0	6	10		1150	75727
0	6	10		1175	75781

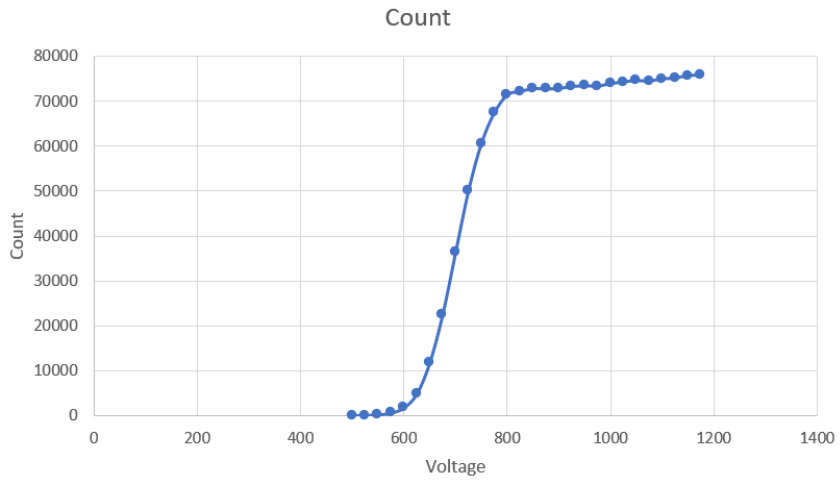


Figure 3: plot of different voltages for α particle

2.1.2 For β particle

Source	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count	Voltage
Ba-131	none	0	6	10	0	500
Ba-131	none	0	6	10	0	525
Ba-131	none	0	6	10	4	550
Ba-131	none	0	6	10	32	575
Ba-131	none	0	6	10	97	600
Ba-131	none	0	6	10	269	625
Ba-131	none	0	6	10	665	650
Ba-131	none	0	6	10	1468	675
Ba-131	none	0	6	10	2792	700
Ba-131	none	0	6	10	4858	725
Ba-131	none	0	6	10	7111	750
Ba-131	none	0	6	10	9386	775
Ba-131	none	0	6	10	10938	800
Ba-131	none	0	6	10	11494	825
Ba-131	none	0	6	10	11398	850
Ba-131	none	0	6	10	11858	875
Ba-131	none	0	6	10	12126	900
Ba-131	none	0	6	10	12040	925
Ba-131	none	0	6	10	12173	950
Ba-131	none	0	6	10	12359	975
Ba-131	none	0	6	10	12486	1000
Ba-131	none	0	6	10	12724	1025
Ba-131	none	0	6	10	12742	1050
Ba-131	none	0	6	10	12773	1075
Ba-131	none	0	6	10	13295	1100
Ba-131	none	0	6	10	13334	1125
Ba-131	none	0	6	10	13709	1150
Ba-131	none	0	6	10	14454	1175
Ba-131	none	0	6	10	15950	1200

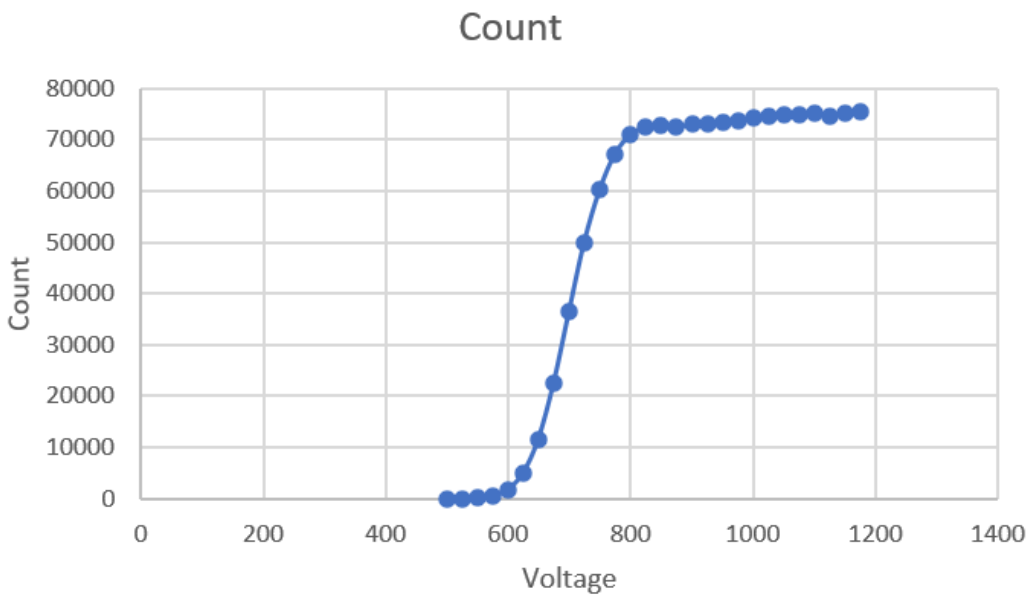


Figure 4: plot of different voltages for β particle

2.1.3 For Gamma Particle

Source	Age (sec)	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count	Voltage
Ba-131		none	0	6	10	0	500
Ba-131		none	0	6	10	0	525
Ba-131		none	0	6	10	4	550
Ba-131		none	0	6	10	32	575
Ba-131		none	0	6	10	97	600
Ba-131		none	0	6	10	269	625
Ba-131		none	0	6	10	665	650
Ba-131		none	0	6	10	1468	675
Ba-131		none	0	6	10	2792	700
Ba-131		none	0	6	10	4858	725
Ba-131		none	0	6	10	7111	750
Ba-131		none	0	6	10	9386	775
Ba-131		none	0	6	10	10938	800
Ba-131		none	0	6	10	11494	825
Ba-131		none	0	6	10	11398	850
Ba-131		none	0	6	10	11858	875
Ba-131		none	0	6	10	12126	900
Ba-131		none	0	6	10	12040	925
Ba-131		none	0	6	10	12173	950
Ba-131		none	0	6	10	12359	975
Ba-131		none	0	6	10	12486	1000
Ba-131		none	0	6	10	12724	1025
Ba-131		none	0	6	10	12742	1050
Ba-131		none	0	6	10	12773	1075
Ba-131		none	0	6	10	13295	1100
Ba-131		none	0	6	10	13334	1125
Ba-131		none	0	6	10	13709	1150
Ba-131		none	0	6	10	14454	1175
Ba-131		none	0	6	10	15950	1200

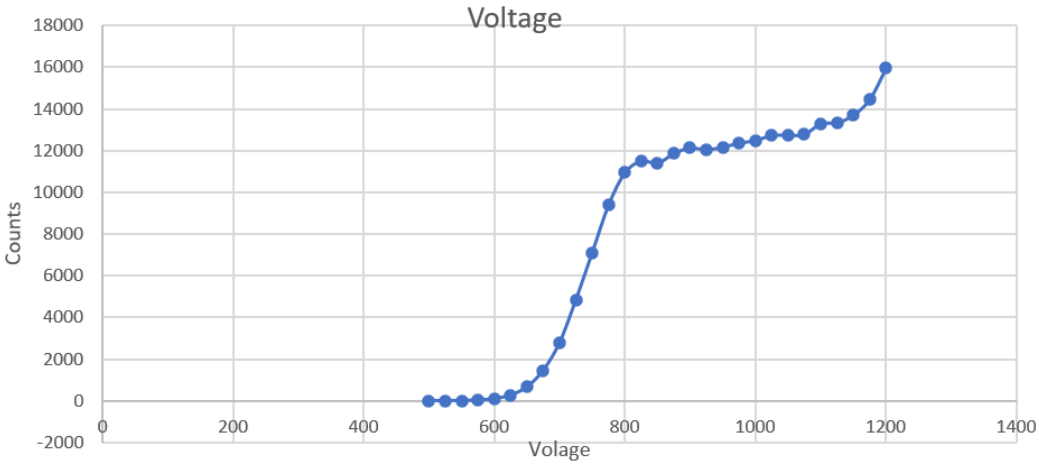


Figure 5: plot of different voltages for γ particle

2.2 Conclusion

From the above mentioned plots it is clear that the plateau region lies between $800V$ to $1000V$.

So for the operating range we will choose the voltage around $1000V$.

3 Stochastic nature of radiation:

Radioactive decay is a random process at the atomic level. Each nucleus of a radioactive substance has a fixed probability of decaying in a given time interval, but when exactly a particular nucleus will decay is unpredictable. This randomness is the essence of the stochastic nature of radiation.

The experiment using a Geiger-Müller (GM) counter to measure radiation demonstrates the stochastic nature of radioactive decay. "Stochastic" refers to processes that are inherently random and unpredictable, though governed by certain probabilities.

3.1 Experiment Overview:

3.1.1 Setup:

- A GM counter is used to detect and count the number of radioactive particles emitted by a radioactive source.
- The GM counter is connected to a counting device, such as a scaler or a computer, that records the number of counts over a set period of time.

3.1.2 Procedure:

- Place the radioactive source near the GM counter.Start the counter and record the number of counts detected in fixed intervals, say every minute or every second.
- Repeat the measurement multiple times for the same time intervals.

3.1.3 Observation:

- The number of counts recorded in each time interval will vary, even if the time interval and distance from the source remain constant.
- This variation in counts reflects the stochastic nature of radioactive decay. Although the average rate of decay (i.e., the mean number of counts per time interval) can be determined, individual measurements will fluctuate around this mean.

3.2 Experimental Data:

Source	**Age (sec)**	**Shield Material**	**Shield Thickness (mm)**	**Distance (cm)**	**Count Duration (sec)**	**Count**
Sr-90		none	0	11	1	4627
Sr-90		none	0	11	1	4465
Sr-90		none	0	11	1	4524
Sr-90		none	0	11	1	4462
Sr-90		none	0	11	1	4554
Sr-90		none	0	11	1	4489
Sr-90		none	0	11	1	4557
Sr-90		none	0	11	1	4547
Sr-90		none	0	11	1	4464
Sr-90		none	0	11	1	4669
Sr-90		none	0	11	1	4547
Sr-90		none	0	11	1	4555
Sr-90		none	0	11	1	4577
Sr-90		none	0	11	1	4580
Sr-90		none	0	11	1	4530

Sr-90		none	0	11	1	4657
Sr-90		none	0	11	1	4556
Sr-90		none	0	11	1	4523
Sr-90		none	0	11	1	4534
Sr-90		none	0	11	1	4463
Sr-90		none	0	11	1	4494
Sr-90		none	0	11	1	4382
Sr-90		none	0	11	1	4535
Sr-90		none	0	11	1	4438
Sr-90		none	0	11	1	4593
Sr-90		none	0	11	1	4473
Sr-90		none	0	11	1	4560
Sr-90		none	0	11	1	4666
Sr-90		none	0	11	1	4506
Sr-90		none	0	11	1	4416
Sr-90		none	0	11	1	4531
Sr-90		none	0	11	1	4461
Sr-90		none	0	11	1	4695
Sr-90		none	0	11	1	4513
Sr-90		none	0	11	1	4397
Sr-90		none	0	11	1	4543
Sr-90		none	0	11	1	4607
Sr-90		none	0	11	1	4667
Sr-90		none	0	11	1	4464
Sr-90		none	0	11	1	4497
Sr-90		none	0	11	1	4558
Sr-90		none	0	11	1	4556
Sr-90		none	0	11	1	4582
Sr-90		none	0	11	1	4418
Sr-90		none	0	11	1	4459
Sr-90		none	0	11	1	4553
Sr-90		none	0	11	1	4429
Sr-90		none	0	11	1	4605
Sr-90		none	0	11	1	4431
Sr-90		none	0	11	1	4523
Sr-90		none	0	11	1	4526
Sr-90		none	0	11	1	4576
Sr-90		none	0	11	1	4415
Sr-90		none	0	11	1	4431
Sr-90		none	0	11	1	4438
Sr-90		none	0	11	1	4497
Sr-90		none	0	11	1	4465
Sr-90		none	0	11	1	4621
Sr-90		none	0	11	1	4582
Sr-90		none	0	11	1	4543
Sr-90		none	0	11	1	4463
Sr-90		none	0	11	1	4630
Sr-90		none	0	11	1	4545
Sr-90		none	0	11	1	4429
Sr-90		none	0	11	1	4546
Sr-90		none	0	11	1	4561
Sr-90		none	0	11	1	4421
Sr-90		none	0	11	1	4559
Sr-90		none	0	11	1	4569
Sr-90		none	0	11	1	4568
Sr-90		none	0	11	1	4468

Sr-90		none	0	11	1	4472
Sr-90		none	0	11	1	4765
Sr-90		none	0	11	1	4499
Sr-90		none	0	11	1	4518
Sr-90		none	0	11	1	4538
Sr-90		none	0	11	1	4514
Sr-90		none	0	11	1	4590
Sr-90		none	0	11	1	4590
Sr-90		none	0	11	1	4505
Sr-90		none	0	11	1	4497
Sr-90		none	0	11	1	4577
Sr-90		none	0	11	1	4601
Sr-90		none	0	11	1	4458
Sr-90		none	0	11	1	4545
Sr-90		none	0	11	1	4572
Sr-90		none	0	11	1	4550
Sr-90		none	0	11	1	4644
Sr-90		none	0	11	1	4525
Sr-90		none	0	11	1	4667
Sr-90		none	0	11	1	4632
Sr-90		none	0	11	1	4551
Sr-90		none	0	11	1	4427
Sr-90		none	0	11	1	4563
Sr-90		none	0	11	1	4496
Sr-90		none	0	11	1	4472
Sr-90		none	0	11	1	4402
Sr-90		none	0	11	1	4609
Sr-90		none	0	11	1	4584
Sr-90		none	0	11	1	4476

3.2.1 Plots:

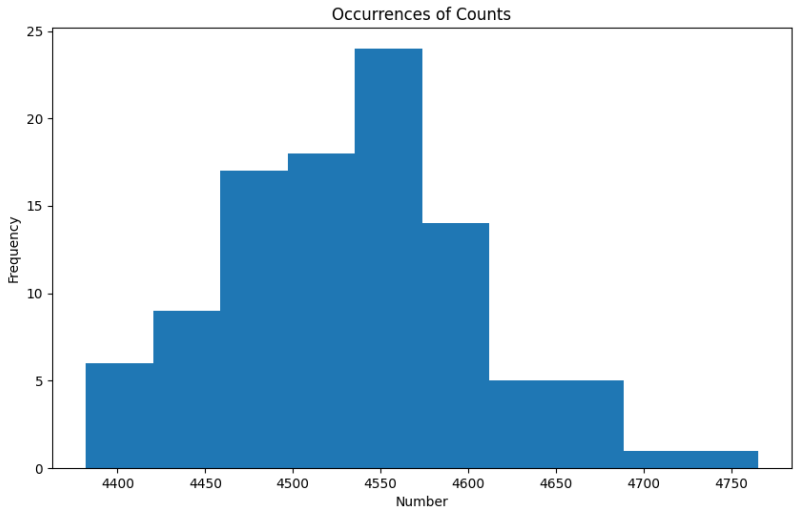


Figure 6: Distributions

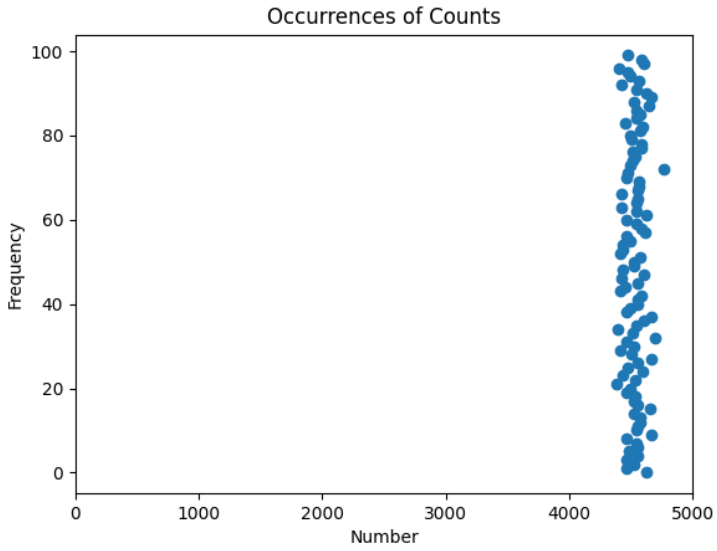


Figure 7: Frequency vs Counts

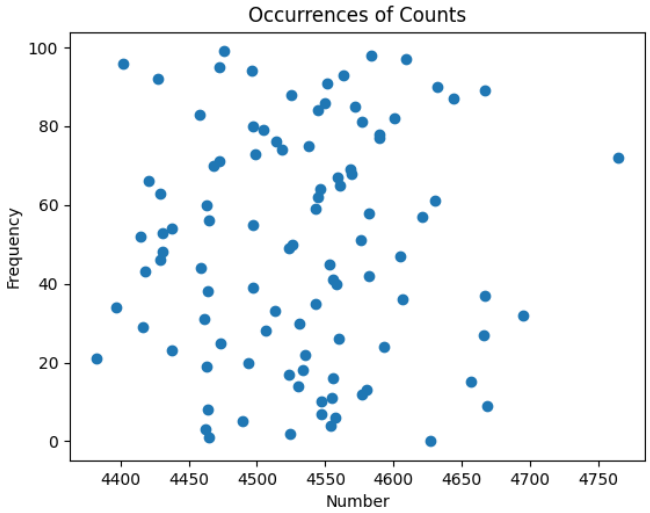


Figure 8: Frequency vs Counts(zoomed-in)

3.2.2 Analysis:

- **Poisson Distribution:** The distribution of counts observed in the experiment typically follows a Poisson distribution, which is characteristic of random processes where events occur independently and with a constant average rate.
- **Mean and Variance:** In a Poisson distribution, the mean and variance are equal. By analyzing the collected data, one can calculate the mean count rate and compare it to the variance to see if they match, further confirming the stochastic nature.

3.3 Conclusion:

The experiment with a GM counter vividly demonstrates the stochastic nature of radioactive decay, as evidenced by the random fluctuations in the count rate. Despite of the randomness, the overall behavior of the system can be statistically predicted and described by probability distributions, such as the Poisson distribution.

4 Range of α, β and γ particles:

The range of radiation types α, β, and γ varies significantly:

- **Alpha Particles:** These are heavy and positively charged. They have a very short range in air, typically a few centimeters. For example, alpha particles from radon decay might travel only about 4–7 cm in air and are stopped by a sheet of paper or even the outer layer of human skin.
- **Beta Particles:** These are lighter and more penetrating than alpha particles. Beta particles can travel several meters in air, with a range dependent on their energy. For instance, beta particles from common isotopes might travel from a few meters up to 10 meters in air. They can be stopped by materials like plastic, glass, or a few millimeters of aluminum.
- **Gamma Rays:** Gamma rays are highly penetrating electromagnetic waves with no mass or charge. They can travel significant distances in air, potentially hundreds of meters, and require dense materials like lead or several centimeters of concrete to be effectively attenuated. The range in air is essentially unlimited for practical purposes.

4.1 Configuration:

Fix the Voltage: Ensure the GM counter is set to its optimal operating voltage (usually specified in the device manual).
Set the Counting Time: Choose a suitable counting period, typically a few minutes, depending on the activity level of the source.

4.2 Experimental Procedure

Measure Background Radiation:

Before placing the radioactive source, measure the background radiation to account for natural radiation. Record this as a baseline count rate.
Place the Radioactive Source:

Position the radioactive source at a specific distance from the GM counter.
Take Measurements:

Start the GM counter and record the number of counts detected over the set counting time. Repeat this measurement several times at each distance to ensure accuracy and account for statistical variation.
Change Distance:

Move the radioactive source to a new distance and repeat the measurements.
Calculate Count Rate:

Subtract the background radiation from each count measurement to get the net count rate.

These ranges are influenced by the energy of the radiation and the density of the medium through which it travels.

4.3 Error Estimation

To estimate the error in the count measurements,

1. Calculate the Average Count Rate: For each distance d, compute the average count rate $\bar{C}_d$ from multiple measurements C_i:

$$\bar{C}_d = \frac{1}{N} \sum_{i=1}^{N} C_i$$

2. Calculate the Standard Deviation: Determine the standard deviation σ_{C_d} for each distance:

$$\sigma_{C_d} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (C_i - \bar{C}_d)^2}$$

3. Calculate the Error: The statistical error ΔC_d is approximated by:

$$\Delta C_d = \frac{\sigma_{C_d}}{\sqrt{N}}$$

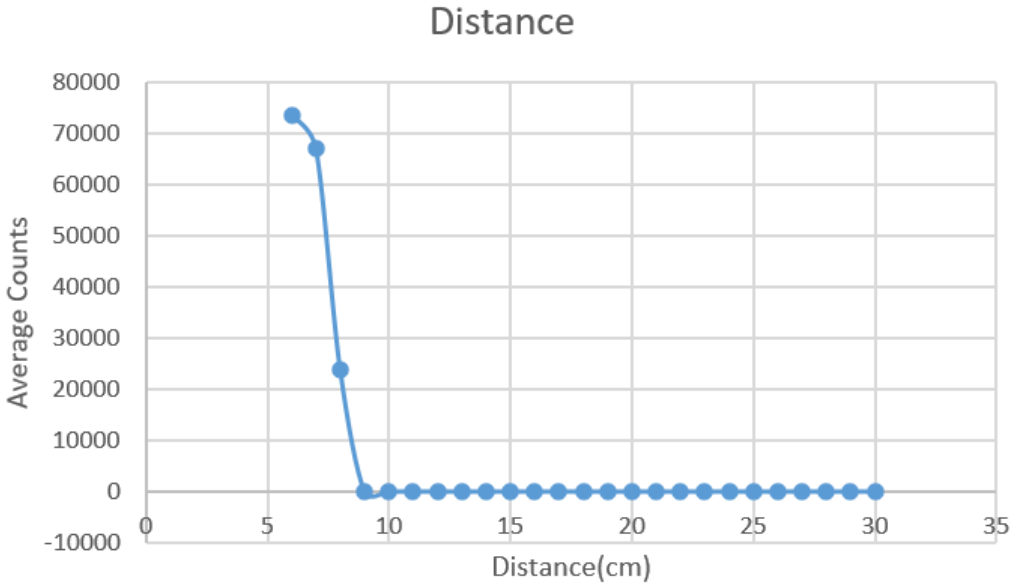
4.4 Experimental data

4.4.1 For α Particle

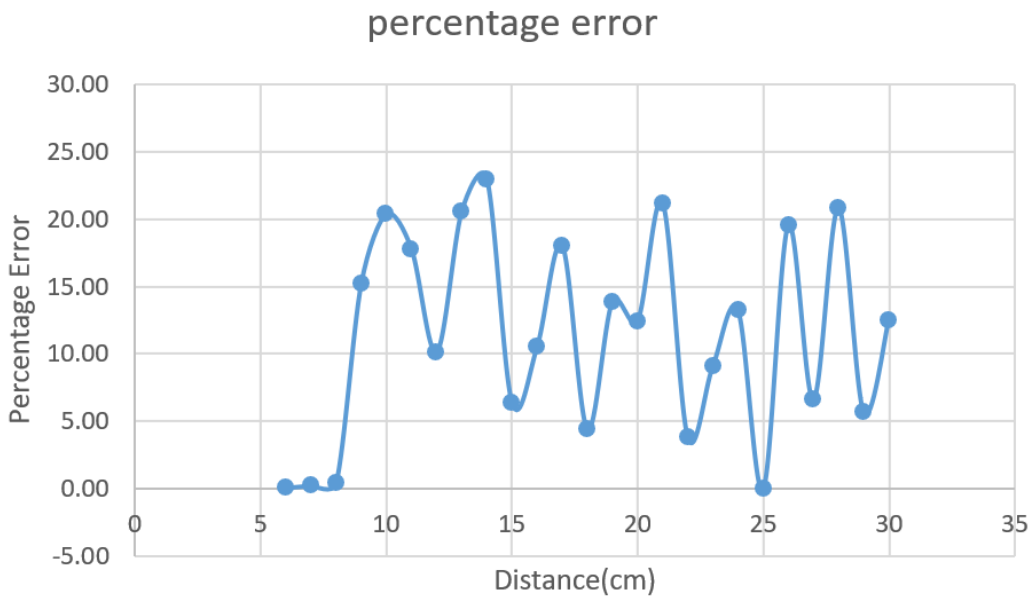
Isotope	**distance**	**Exposure**	**Counts**	**Average Count**	**Distance**	**error**	**percentage error**
Am-242	6	10	73731	73600.00	6	71.93	0.10
Am-242	6	10	73483	67063.33	7	175.12	0.26
Am-242	6	10	73586	23848.67	8	109.79	0.46
Am-242	7	10	67198	13.33	9	2.03	15.21
Am-242	7	10	66716	13.00	10	2.65	20.35
Am-242	7	10	67276	13.00	11	2.31	17.76
Am-242	8	10	23634	14.33	12	1.45	10.14
Am-242	8	10	23916	12.67	13	2.60	20.55
Am-242	8	10	23996	14.00	14	3.21	22.96
Am-242	9	10	17	9.00	15	0.58	6.42
Am-242	9	10	10	12.67	16	1.33	10.53
Am-242	9	10	13	13.33	17	2.40	18.03
Am-242	10	10	17	13.00	18	0.58	4.44
Am-242	10	10	8	11.00	19	1.53	13.89
Am-242	10	10	14	9.67	20	1.20	12.43
Am-242	11	10	17	8.33	21	1.76	21.17
Am-242	11	10	9	8.67	22	0.33	3.85
Am-242	11	10	13	9.67	23	0.88	9.12
Am-242	12	10	14	13.33	24	1.76	13.23
Am-242	12	10	17	9.00	25	0.00	0.00
Am-242	12	10	12	13.67	26	2.67	19.51
Am-242	13	10	17	13.33	27	0.88	6.61
Am-242	13	10	13	10.00	28	2.08	20.82
Am-242	13	10	8	11.67	29	0.67	5.71
Am-242	14	10	20	13.33	30	1.67	12.50
Am-242	14	10	9				
Am-242	14	10	13				
Am-242	15	10	10				
Am-242	15	10	9				
Am-242	15	10	8				
Am-242	16	10	10				
Am-242	16	10	14				
Am-242	16	10	14				
Am-242	17	10	12				
Am-242	17	10	18				
Am-242	17	10	10				
Am-242	18	10	13				
Am-242	18	10	12				
Am-242	18	10	14				
Am-242	19	10	13				
Am-242	19	10	8				
Am-242	19	10	12				
Am-242	20	10	12				
Am-242	20	10	9				
Am-242	20	10	8				
Am-242	21	10	9				
Am-242	21	10	11				

Am-242	21	10	5				
Am-242	22	10	8				
Am-242	22	10	9				
Am-242	22	10	9				
Am-242	23	10	11				
Am-242	23	10	8				
Am-242	23	10	10				
Am-242	24	10	10				
Am-242	24	10	16				
Am-242	24	10	14				
Am-242	25	10	9				
Am-242	25	10	9				
Am-242	25	10	9				
Am-242	26	10	19				
Am-242	26	10	11				
Am-242	26	10	11				
Am-242	27	10	15				
Am-242	27	10	12				
Am-242	27	10	13				
Am-242	28	10	9				
Am-242	28	10	14				
Am-242	28	10	7				
Am-242	29	10	13				
Am-242	29	10	11				
Am-242	29	10	11				
Am-242	30	10	15				
Am-242	30	10	10				
Am-242	30	10	15				

Plot:



Error Plot:



4.4.2 Conclusion:

By this plots this is clear that α particles have very low range and it is clearly visible by large error values at higher distances and also matching of average counts with the counts of environment when there is no radio isotopes are present.

4.4.3 For β Particle:

Source	Distance (cm)	Count Duration (sec)	Count	Distance	Avg Count	Error	percentage Error
Sr-90	6	10	73336	6	73652.67	235.10	0.32
Sr-90	6	10	73510	7	67070.00	144.58	0.22
Sr-90	6	10	74112	8	60899.67	112.57	0.18
Sr-90	7	10	67287	9	55244.00	76.85	0.14
Sr-90	7	10	66796	10	49635.67	74.55	0.15
Sr-90	7	10	67127	11	45134.67	156.90	0.35
Sr-90	8	10	61070	12	41139.00	142.86	0.35
Sr-90	8	10	60942	13	37250.33	65.63	0.18
Sr-90	8	10	60687	14	33803.67	99.11	0.29
Sr-90	9	10	55394	15	30966.00	54.22	0.18
Sr-90	9	10	55140	16	28100.00	93.40	0.33
Sr-90	9	10	55198	17	25682.67	54.49	0.21
Sr-90	10	10	49768	18	23553.67	47.81	0.20
Sr-90	10	10	49510	19	21736.00	77.95	0.36
Sr-90	10	10	49629	20	19995.00	131.84	0.66
Sr-90	11	10	45158	21	18447.33	64.72	0.35
Sr-90	11	10	44852	22	17118.00	42.33	0.25
Sr-90	11	10	45394	23	15748.33	82.33	0.52
Sr-90	12	10	40880	24	14800.67	30.01	0.20
Sr-90	12	10	41164	25	13773.67	47.74	0.35
Sr-90	12	10	41373	26	12954.33	18.84	0.15

Sr-90	13	10	37120	27	12050.33	64.59	0.54
Sr-90	13	10	37329	28	11377.67	11.57	0.10
Sr-90	13	10	37302	29	10646.33	33.77	0.32
Sr-90	14	10	33874	30	10045.33	103.14	1.03
Sr-90	14	10	33929				
Sr-90	14	10	33608				
Sr-90	15	10	30937				
Sr-90	15	10	30890				
Sr-90	15	10	31071				
Sr-90	16	10	27976				
Sr-90	16	10	28283				
Sr-90	16	10	28041				
Sr-90	17	10	25601				
Sr-90	17	10	25786				
Sr-90	17	10	25661				
Sr-90	18	10	23647				
Sr-90	18	10	23525				
Sr-90	18	10	23489				
Sr-90	19	10	21690				
Sr-90	19	10	21888				
Sr-90	19	10	21630				
Sr-90	20	10	19734				
Sr-90	20	10	20158				
Sr-90	20	10	20093				
Sr-90	21	10	18338				
Sr-90	21	10	18442				
Sr-90	21	10	18562				
Sr-90	22	10	17198				
Sr-90	22	10	17102				
Sr-90	22	10	17054				
Sr-90	23	10	15904				
Sr-90	23	10	15624				
Sr-90	23	10	15717				
Sr-90	24	10	14742				
Sr-90	24	10	14819				
Sr-90	24	10	14841				
Sr-90	25	10	13688				
Sr-90	25	10	13780				
Sr-90	25	10	13853				
Sr-90	26	10	12992				
Sr-90	26	10	12935				
Sr-90	26	10	12936				
Sr-90	27	10	11965				
Sr-90	27	10	12177				
Sr-90	27	10	12009				
Sr-90	28	10	11393				
Sr-90	28	10	11385				
Sr-90	28	10	11355				
Sr-90	29	10	10595				
Sr-90	29	10	10634				
Sr-90	29	10	10710				
Sr-90	30	10	10228				
Sr-90	30	10	10037				
Sr-90	30	10	9871				

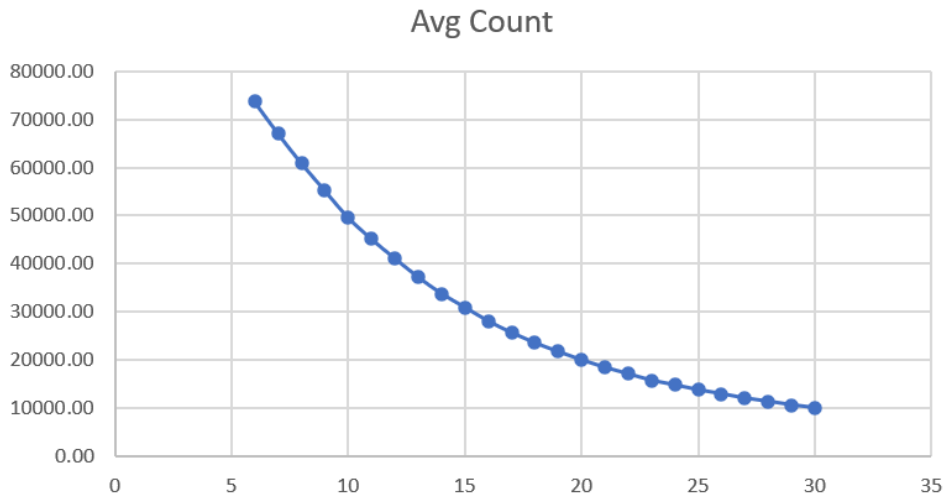
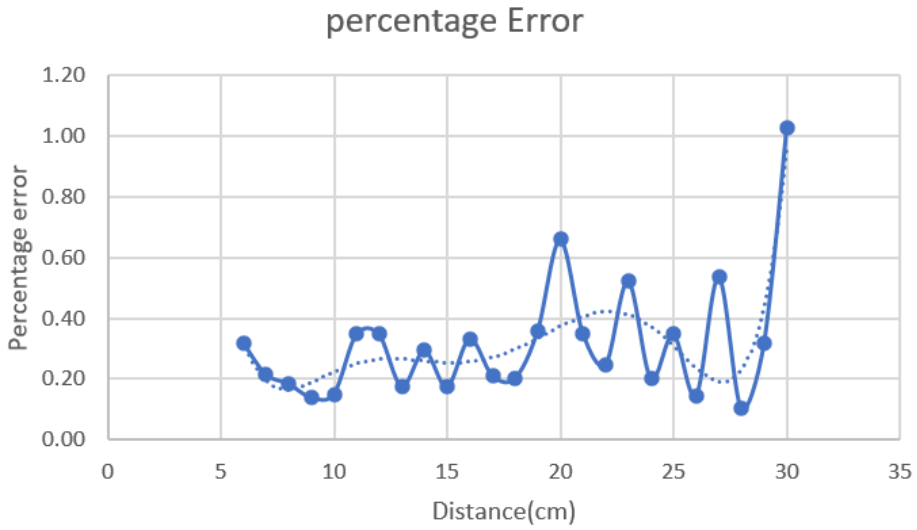


Figure 9: Average Counts vs Distance



4.4.4 Conclusion:

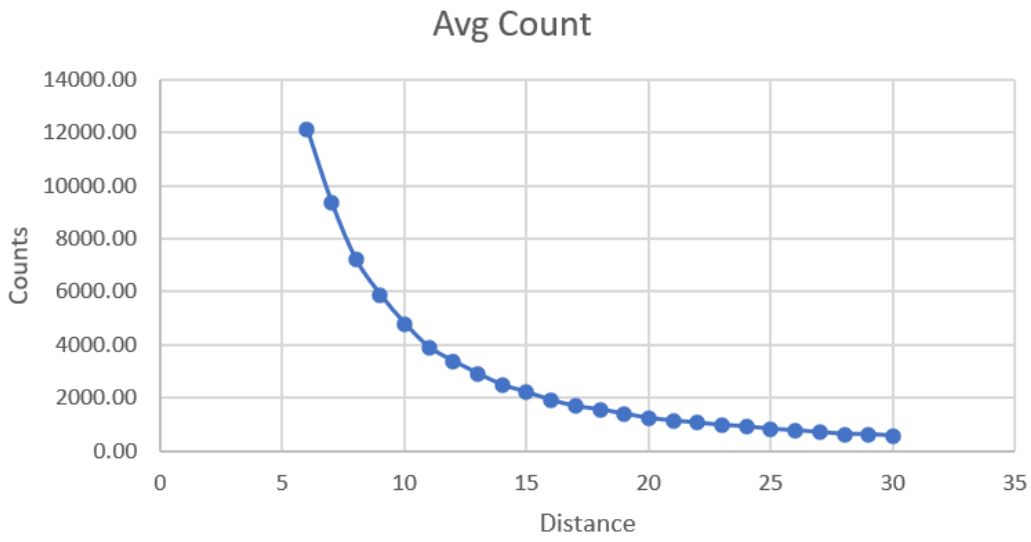
β has moderate range and we can see how count and error is changing with distance.

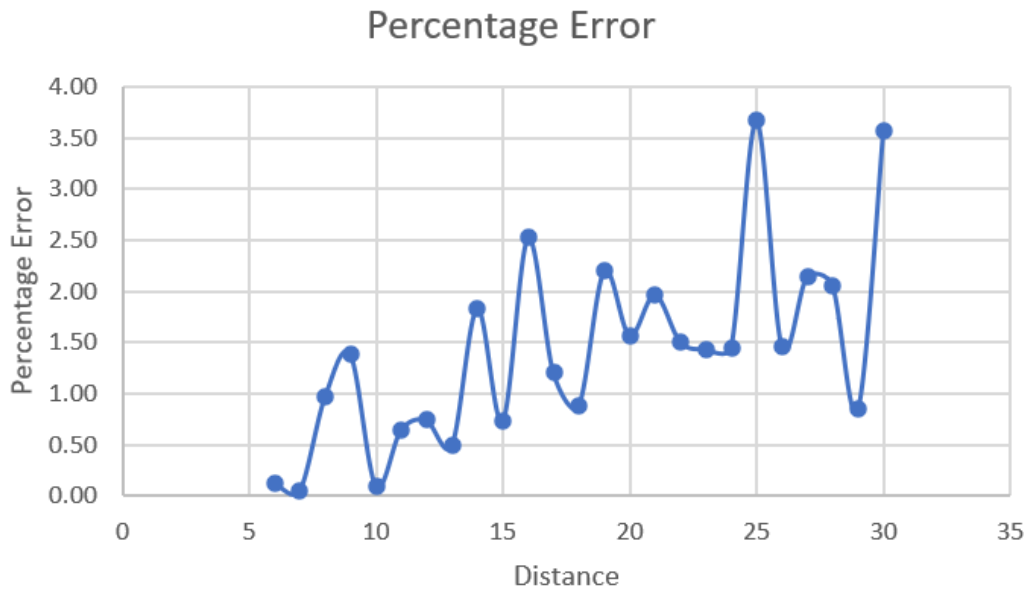
4.4.5 For γ Particle:

Source	**Distance (cm)**	**Count Duration (sec)**	**Count**	**Distance**	**Avg Count**	**error**	**Percentage Error**
Ba-131	6	10	12121	6	12120.67	14.72	0.12
Ba-131	6	10	12095	7	9366.67	4.63	0.05
Ba-131	6	10	12146	8	7213.33	69.96	0.97
Ba-131	7	10	9359	9	5884.33	81.23	1.38
Ba-131	7	10	9366	10	4794.00	4.73	0.10
Ba-131	7	10	9375	11	3901.33	25.20	0.65
Ba-131	8	10	7171	12	3377.00	25.00	0.74
Ba-131	8	10	7350	13	2910.67	14.40	0.49
Ba-131	8	10	7119	14	2480.00	45.57	1.84

Ba-131	9	10	5726	15	2210.00	16.17	0.73
Ba-131	9	10	5995	16	1904.67	48.07	2.52
Ba-131	9	10	5932	17	1679.67	20.19	1.20
Ba-131	10	10	4803	18	1552.33	13.72	0.88
Ba-131	10	10	4792	19	1393.33	30.67	2.20
Ba-131	10	10	4787	20	1231.33	19.34	1.57
Ba-131	11	10	3910	21	1131.00	22.28	1.97
Ba-131	11	10	3940	22	1054.67	15.90	1.51
Ba-131	11	10	3854	23	964.67	13.84	1.43
Ba-131	12	10	3352	24	917.33	13.32	1.45
Ba-131	12	10	3352	25	818.33	30.05	3.67
Ba-131	12	10	3427	26	769.00	11.27	1.47
Ba-131	13	10	2892	27	698.67	15.01	2.15
Ba-131	13	10	2939	28	630.67	12.99	2.06
Ba-131	13	10	2901	29	621.33	5.24	0.84
Ba-131	14	10	2395	30	559.33	19.95	3.57
Ba-131	14	10	2494				
Ba-131	14	10	2551				
Ba-131	15	10	2238				
Ba-131	15	10	2210				
Ba-131	15	10	2182				
Ba-131	16	10	1888				
Ba-131	16	10	1995				
Ba-131	16	10	1831				
Ba-131	17	10	1706				
Ba-131	17	10	1693				
Ba-131	17	10	1640				
Ba-131	18	10	1568				
Ba-131	18	10	1564				
Ba-131	18	10	1525				
Ba-131	19	10	1423				
Ba-131	19	10	1425				
Ba-131	19	10	1332				
Ba-131	20	10	1195				
Ba-131	20	10	1261				
Ba-131	20	10	1238				
Ba-131	21	10	1171				
Ba-131	21	10	1094				
Ba-131	21	10	1128				
Ba-131	22	10	1068				
Ba-131	22	10	1073				
Ba-131	22	10	1023				
Ba-131	23	10	937				
Ba-131	23	10	978				
Ba-131	23	10	979				
Ba-131	24	10	891				
Ba-131	24	10	934				
Ba-131	24	10	927				
Ba-131	25	10	760				
Ba-131	25	10	860				
Ba-131	25	10	835				
Ba-131	26	10	770				
Ba-131	26	10	749				
Ba-131	26	10	788				
Ba-131	27	10	698				
Ba-131	27	10	673				

Ba-131	27	10	725				
Ba-131	28	10	631				
Ba-131	28	10	608				
Ba-131	28	10	653				
Ba-131	29	10	625				
Ba-131	29	10	628				
Ba-131	29	10	611				
Ba-131	30	10	595				
Ba-131	30	10	526				
Ba-131	30	10	557				





4.4.6 Conclusion:

So by the slope of the plot we see gamma has the highest range in all of them.

5 Shielding Effectiveness of Different Materials:

- **Setup:**

  Fix the operating voltage, counting time, and distance between the detector and the radioactive source. Use shields of the same thickness but made from different materials (e.g., paper, aluminum, lead, etc.).

- **Procedure:**

  Measure the background radiation (without any source or shielding). Place the radioactive source at the fixed distance from the detector and record the counts without any shield. Place different shields (one at a time) between the source and the detector and record the counts. Repeat the measurements several times for each material to get an average count.

- **Observations:**

  The material with the least shielding power will result in counts closer to the unshielded counts. The material with the best shielding power will show a significant reduction in counts compared to the unshielded case.

- **Expected Results:**

  Paper: Typically, paper is effective only against alpha particles.

  Aluminum: Effective against alpha and some beta particles.

  Lead: Highly effective against gamma rays and x-rays, and also beta particles.

5.1 Effectiveness of Paper in Shielding Different Particles:

- **Setup:**

  Place the detector very close to the source to maximize count rates.

  Procedure:

  Measure the background radiation. Measure counts without any shield. Place a paper shield between the source and detector and record the counts. Repeat the measurements to estimate the average counts.

  Observations:

  Paper should effectively shield alpha particles, but beta particles and gamma rays may not be significantly attenuated.

- **Expected Results:**

  Alpha particles: Significant reduction in counts.

  Beta particles: Minor reduction in counts.

  Gamma rays: Minimal or no reduction in counts.

5.2 Experimental Data

5.2.1 α particle and different materials

Source	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count(avg)
Am-242	plastic	5	6	10	13
Am-242	copper	5	6	10	16
Am-242	wood	5	6	10	16
Am-242	aluminium	5	6	10	11
Am-242	lead	5	6	10	10
Am-242	glass	5	6	10	13
Am-242	paper	0.2	6	10	299
Am-242	stone	5	6	10	10

5.2.2 β particle and different materials

Source	Sheild Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count(Avg)
Sr-90	plastic	5	6	10	457.6666667
Sr-90	Copper	5	6	10	12.33333333
Sr-90	Wood	5	6	10	889.6666667
Sr-90	Aluminium	5	6	10	14.66666667
Sr-90	Lead	5	6	10	12
Sr-90	Glass	5	6	10	13.66666667
Sr-90	Paper	0.2	6	10	67169
Sr-90	Stone	5	6	10	13.33333333

5.2.3 γ particle and different materials

Source	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count(Avg)
Ba-131	plastic	5	6	10	11511.67
Ba-131	Copper	5	6	10	7220.33
Ba-131	Wood	5	6	10	11650.67
Ba-131	Aluminium	5	6	10	10191.00
Ba-131	Lead	5	6	10	5257.33
Ba-131	Glass	5	6	10	10639.33
Ba-131	Paper	0.2	6	10	12157.33
Ba-131	Stone	5	6	10	10668.00

5.3 Observations:

From all of the data it is clear that Lead has the best shielding power and paper has the least.

5.3.1 Obervation for paper:

As it has a very low thickness eg. $0.2mm$, and it's atomic packing is also not so dense and because of having long chain molecules it's shielding effect is least.

5.4 Applications of Shields:

Medical Radiography: Lead aprons are used to protect patients and staff from unnecessary exposure to x-rays.
Nuclear Power Plants: Thick concrete walls and lead shields are used to protect workers and the environment from radiation.
Radiation Therapy: Shields are used to protect surrounding healthy tissues while targeting tumors with radiation.
Spacecraft: Shields protect astronauts from cosmic radiation and solar flares. Industrial Radiography: Shielding is used in gamma-ray and x-ray inspection to protect operators.

6 Determination of Absorption Coefficient of Different Shielding Materials

In this experiment, the goal is to determine the absorption coefficient (μ) of different shielding materials such as iron (Fe), copper (Cu), and lead (Pb). The absorption coefficient quantifies how effectively a material can attenuate radiation. By measuring the number of counts detected by a Geiger-Müller counter for different thicknesses of shielding materials, and applying an exponential decay model, we can estimate μ for each material.

6.1 Experimental Setup

Fixed Parameters

- **Voltage**: The operating voltage of the Geiger-Müller counter is fixed throughout the experiment.
- **Counting Time**: The counting time is kept constant and long enough to ensure statistically significant measurements.
- **Distance**: The distance between the radioactive source and the detector is kept constant.

Materials

The shielding materials used in this experiment are iron (Fe), copper (Cu), and lead (Pb) with varying thicknesses.

6.2 Procedure

1. **Background Measurement**: Record the background radiation without the source. Repeat the measurement several times to obtain an average.
2. **Unshielded Measurement**: Place the radioactive source at a fixed distance from the detector and record the number of counts without any shielding material. Repeat this measurement several times to get an average count rate, N_0.
3. **Shielded Measurements**:
   - Place a thin layer of the shielding material between the source and the detector.
   - Record the number of counts for that thickness.
   - Repeat the measurement for multiple thicknesses of the same material.
   - For each thickness, take multiple readings to obtain an average count, $N(x)$.
4. **Plotting the Data**: Plot the number of counts ($N(x)$) versus the thickness of the shielding material (x). The relationship is expected to follow the exponential decay model:
$$N(x) = N_0 \cdot e^{-\mu x}$$
5. **Determining the Absorption Coefficient**:
   - Linearize the equation by taking the natural logarithm of both sides:
$$\ln(N(x)) = \ln(N_0) - \mu x$$
   - Plot $\ln(N(x))$ versus x. The slope of the line will be $-\mu$.

6.3 Error Analysis

Uncertainty in Count Measurements

The standard deviation of the counts for each thickness is used to estimate the uncertainty in the count measurements:

$$\sigma_{N(x)} = \sqrt{\frac{\sum_{i=1}^{n}(N_i(x) - \overline{N(x)})^2}{n-1}}$$

where $N_i(x)$ are individual count measurements for a given thickness x, $\overline{N(x)}$ is the average count, and n is the number of measurements.

Uncertainty in the Logarithmic Counts

The uncertainty in the logarithm of the counts, $\ln(N(x))$, can be estimated using:

$$\sigma_{\ln(N(x))} = \frac{\sigma_{N(x)}}{N(x)}$$

Uncertainty in the Absorption Coefficient

The absorption coefficient μ is obtained from the slope of the linearized plot. The uncertainty in μ can be estimated from the standard error of the slope, σ_μ, which is given by:

$$\sigma_\mu = \sqrt{\frac{1}{n-2}\left(\frac{\sum_{i=1}^{n}(\ln(N(x_i)) - (a - \mu x_i))^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}\right)}$$

where x_i is the thickness, $\overline{x}$ is the mean thickness, a is the intercept of the linearized plot, and n is the number of data points.

6.4 Experimental data for Cu:

6.4.1 For α particle

Source	Count	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count after subtracting surrounding noise	Thickness	avg Count	error in Count	error (%)
Am-242	24	copper	0.5	6	10	14	0.5	10.67	2.03	19.01
Am-242	17	copper	0.5	6	10	7	1	7.67	1.20	15.68
Am-242	21	copper	0.5	6	10	11	2	7.33	1.86	25.31
Am-242	16	copper	1	6	10	6	3	4.00	1.53	38.19
Am-242	17	copper	1	6	10	7	4	8.00	1.73	21.65
Am-242	20	copper	1	6	10	10	6	7.33	0.88	12.03
Am-242	15	copper	2	6	10	5	8	9.00	0.58	6.42
Am-242	16	copper	2	6	10	6	10	9.67	2.40	24.87
Am-242	21	copper	2	6	10	11				
Am-242	11	copper	3	6	10	1				
Am-242	15	copper	3	6	10	5				
Am-242	16	copper	3	6	10	6				
Am-242	15	copper	4	6	10	5				
Am-242	21	copper	4	6	10	11				
Am-242	18	copper	4	6	10	8				
Am-242	16	copper	6	6	10	6				
Am-242	17	copper	6	6	10	7				

Am-242	19	copper	6	6	10	9				
Am-242	20	copper	8	6	10	10				
Am-242	18	copper	8	6	10	8				
Am-242	19	copper	8	6	10	9				
Am-242	23	copper	10	6	10	13				
Am-242	15	copper	10	6	10	5				
Am-242	21	copper	10	6	10	11				

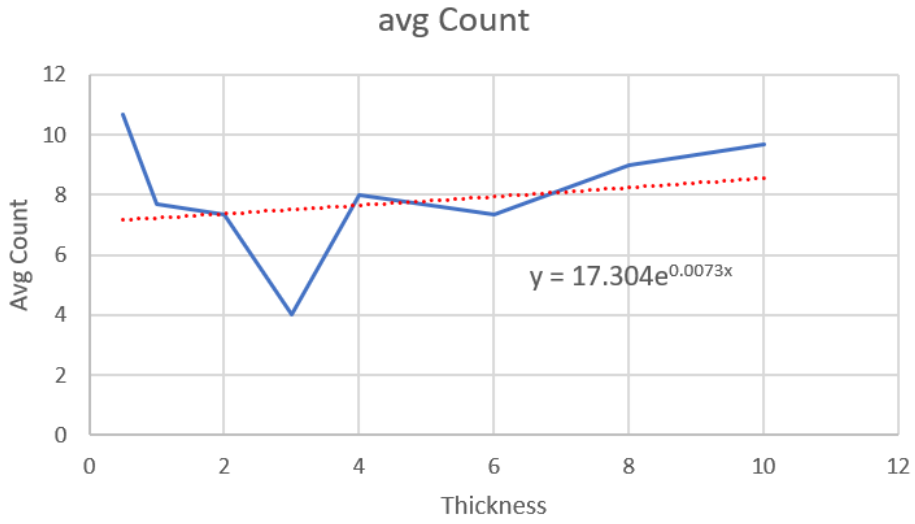


Figure 10: Plot for all thickness

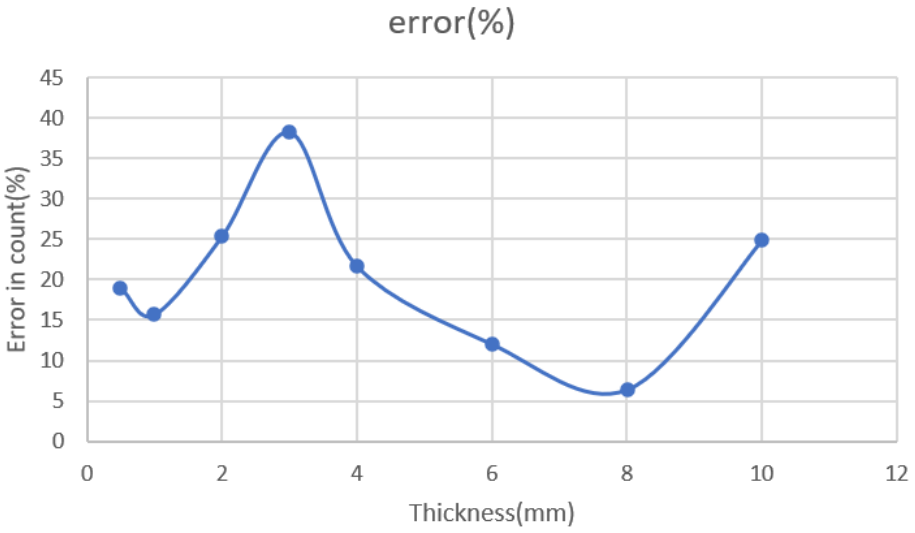


Figure 11: Error in Counts

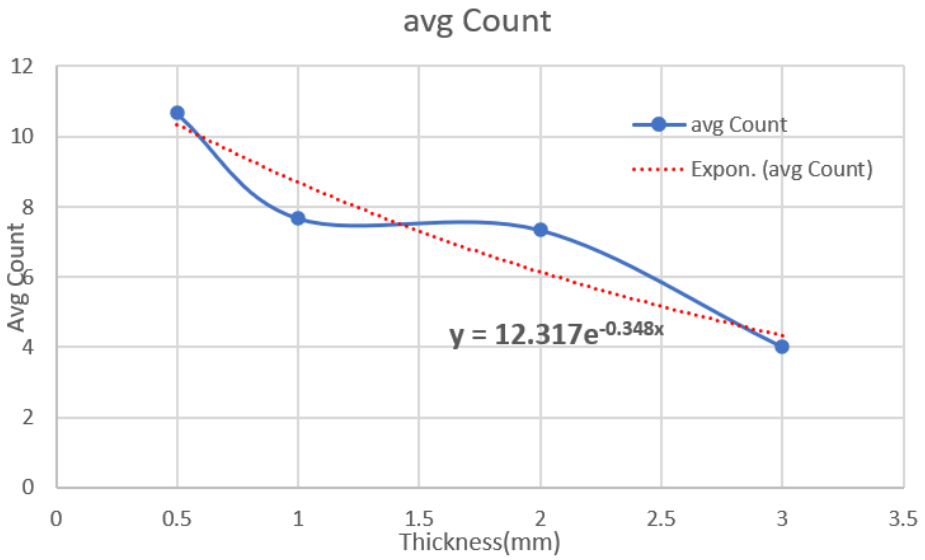


Figure 12: Plot for initial thicknesses

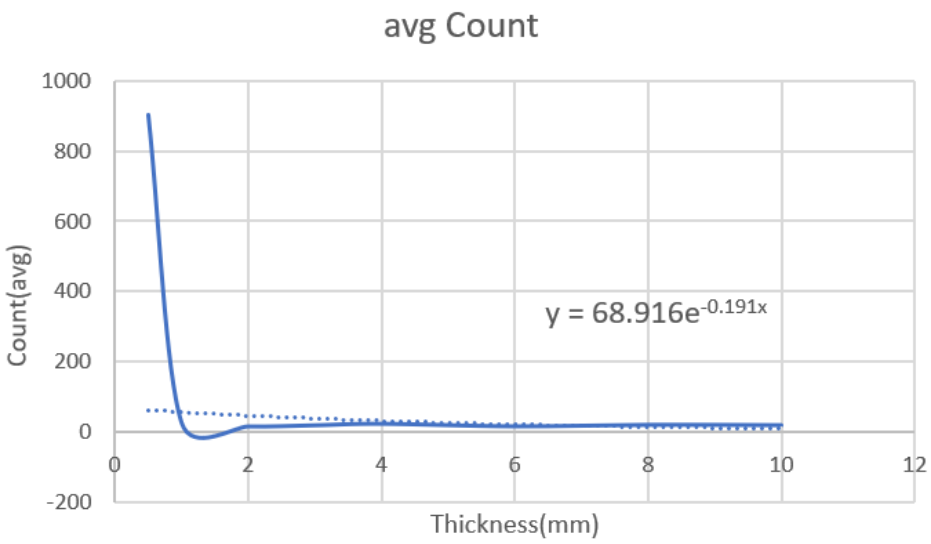
Discussion:

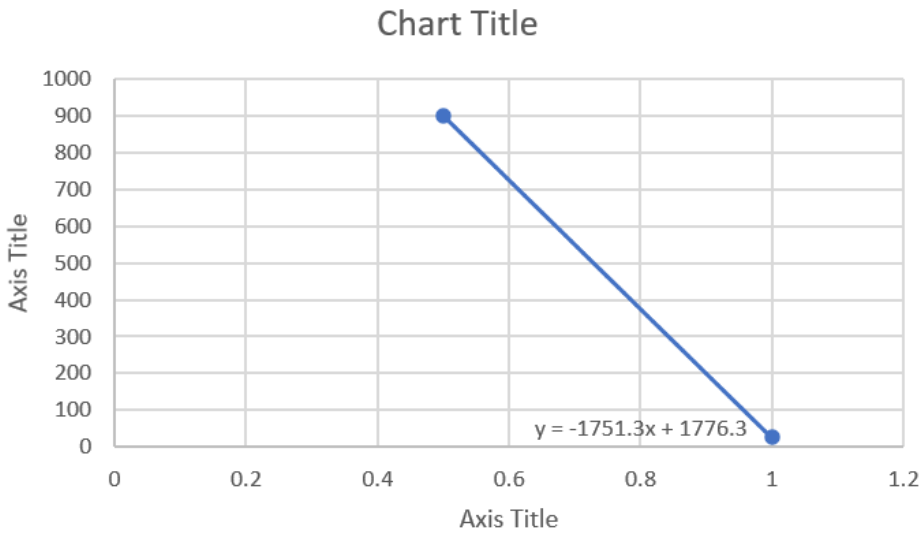
From the Previous experiments it is clear that alpha particle has very low penetration power, thats why after a certain thickness the counts matches with the environmental noise counts and because of this we can't get the plot for further thicknesses. ***But still the value of absorption coefficient value is too high that curve get flat in the beginning of 0.5mm plate and matches the environmental noise counts.***

such as, $\lim_{\mu\to\infty} e^{-\mu x} \approx 0$**.**

6.4.2 For β particle:

Source	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count	Thickness	avg Count	error	Error(%)
Sr-90	copper	0.5	6	10	882	0.5	900.6666667	9.351173426	1.038250195
Sr-90	copper	0.5	6	10	909	1	25	2.309401077	9.237604307
Sr-90	copper	0.5	6	10	911	2	14.66666667	1.333333333	9.090909091
Sr-90	copper	1	6	10	29	3	18	1.732050808	9.622504486
Sr-90	copper	1	6	10	25	4	22.33333333	1.452966315	6.505819319
Sr-90	copper	1	6	10	21	6	15	3.605551275	24.0370085
Sr-90	copper	2	6	10	16	8	19.33333333	1.452966315	7.515343006
Sr-90	copper	2	6	10	12	10	18.33333333	1.333333333	7.272727273
Sr-90	copper	2	6	10	16				
Sr-90	copper	3	6	10	18				
Sr-90	copper	3	6	10	21				
Sr-90	copper	3	6	10	15				
Sr-90	copper	4	6	10	22				
Sr-90	copper	4	6	10	25				
Sr-90	copper	4	6	10	20				
Sr-90	copper	6	6	10	10				
Sr-90	copper	6	6	10	22				
Sr-90	copper	6	6	10	13				
Sr-90	copper	8	6	10	19				
Sr-90	copper	8	6	10	22				
Sr-90	copper	8	6	10	17				
Sr-90	copper	10	6	10	17				
Sr-90	copper	10	6	10	21				
Sr-90	copper	10	6	10	17				





6.4.3 Discussion:

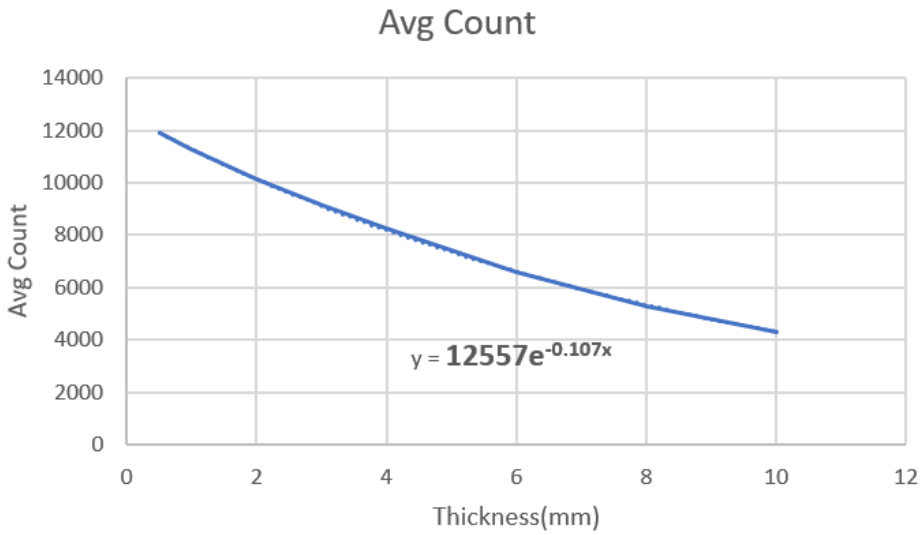
Here is also because of lower penetration ability of β particle plot flats very fast.

But we know for small values of x, $e^{-\mu x} \approx 1 - \mu x$.

So, we have plotted the graph for small x and from that we are getting the slope of **1751.3/1776.3 = 0.986 ≈ μ**(absorption coefficient).

6.4.4 For γ particle:

Source	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count	Thickness	Avg Count
Ba-131	copper	0.5	6	10	11897	0.5	11877
Ba-131	copper	0.5	6	10	11762	1	11263
Ba-131	copper	0.5	6	10	11972	2	10118.66667
Ba-131	copper	1	6	10	11131	3	9152
Ba-131	copper	1	6	10	11433	4	8233
Ba-131	copper	1	6	10	11225	6	6598.666667
Ba-131	copper	2	6	10	10155	8	5315.333333
Ba-131	copper	2	6	10	10135	10	4313.333333
Ba-131	copper	2	6	10	10066		
Ba-131	copper	3	6	10	9106		
Ba-131	copper	3	6	10	9142		
Ba-131	copper	3	6	10	9208		
Ba-131	copper	4	6	10	8220		
Ba-131	copper	4	6	10	8242		
Ba-131	copper	4	6	10	8237		
Ba-131	copper	6	6	10	6578		
Ba-131	copper	6	6	10	6578		
Ba-131	copper	6	6	10	6640		
Ba-131	copper	8	6	10	5353		
Ba-131	copper	8	6	10	5344		
Ba-131	copper	8	6	10	5249		
Ba-131	copper	10	6	10	4408		
Ba-131	copper	10	6	10	4288		
Ba-131	copper	10	6	10	4244		



6.4.5 Discussion:

From the above plot we can say the Absorption Coefficient is $0.107 \boldsymbol{mm^{-1}}$

6.5 Results and Discussion

Expected Results

- **Iron (Fe)**: Expected to have a moderate absorption coefficient, effective for gamma and x-rays.
- **Copper (Cu)**: Similar to iron, with a slightly higher absorption coefficient due to higher density.
- **Lead (Pb)**: Expected to have the highest absorption coefficient, most effective in shielding against gamma rays.

6.6 Applications

- **Radiation Shielding Design**: Knowledge of μ helps in designing appropriate shields in medical, industrial, and nuclear applications.
- **Material Selection**: Helps in choosing the most effective material for specific types of radiation and conditions.

7 Estimating the Half-Life of Protactinium-234

To estimate the half-life of Protactinium-234 (^{234}Pa), you would typically follow these steps:

7.1 Set Up the Experiment

- **Operating Voltage**: Set the Geiger-Müller counter to the appropriate operating voltage to ensure it can detect the decay particles (typically beta particles in the case of ^{234}Pa).
- **Distance of the Detector from the Source**: Position the detector at a fixed distance from the ^{234}Pa source to ensure a consistent count rate. The distance should be such that the detector can capture a significant number of decay events without being saturated by the count rate.
- **Counting Interval**: Choose a suitable counting interval (e.g., every 10 seconds, 1 minute, etc.) to record the number of decays over time. The interval should be short enough to capture the decay curve but long enough to get a statistically significant number of counts.

7.2 Determine the Half-Life

The half-life $t_{1/2}$ is the time required for the count rate to decrease to half its initial value. This can be determined directly from the plot or by fitting the data to the exponential decay equation:

$$N(t) = N_0 e^{-\lambda t}$$

where:

- $N(t)$ is the count rate at time t.
- N_0 is the initial count rate.
- λ is the decay constant, related to the half-life by $t_{1/2} = \frac{\ln(2)}{\lambda}$.

8 Feasibility and Considerations

8.1 Half-Life of Protactinium-234

^{234}Pa has a half-life of approximately 1.17 minutes (about 70 seconds). This is relatively short, which makes it feasible to observe the decay process in real-time within a laboratory setting.

8.2 Feasibility

8.2.1 Positive Aspects

- **Short Experiment Duration**: The short half-life allows the entire decay process to be observed within a reasonable time frame, typically within 10–20 minutes.
- **Clear Decay Curve**: Given the short half-life, the decay curve will be well-defined, making it easier to accurately estimate the half-life.

8.2.2 Challenges

- **High Activity**: Due to the short half-life, ^{234}Pa can exhibit high activity initially, which could saturate the detector or require the experiment to be performed quickly.
- **Background Radiation**: If the background radiation is not properly accounted for, it could interfere with the measurements, especially if the detector is too far from the source or if the source's activity decreases significantly over the course of the experiment.
- **Detector Efficiency**: The detector must be efficient enough to detect the beta particles emitted by ^{234}Pa. If the detector's efficiency is low, the results could be skewed, leading to an inaccurate estimate of the half-life.

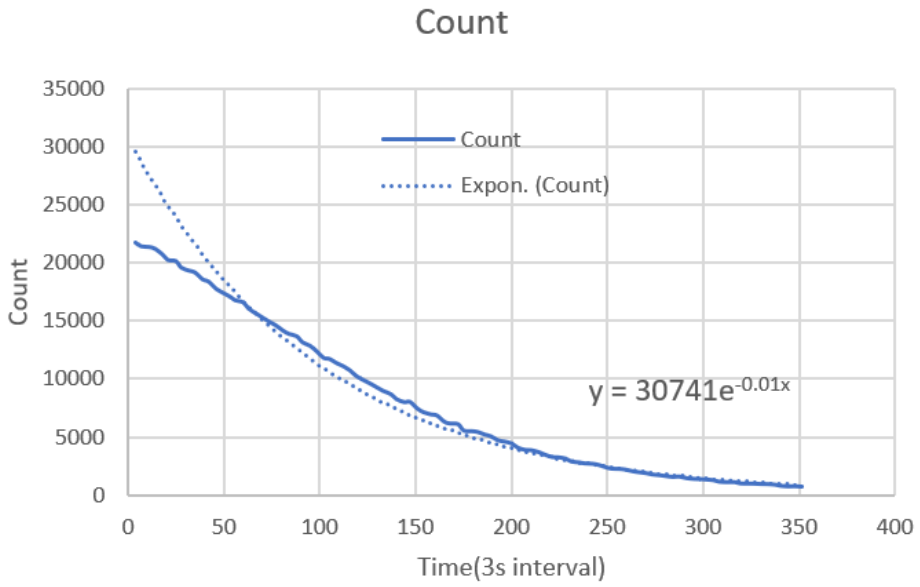
Using Protactinium-234 to carry out a half-life experiment is possible and practical due to its relatively short half-life, which allows for real-time observation of the decay process. However, the experiment requires careful consideration of the operating voltage, detector distance, and counting interval to ensure accurate results. Challenges such as high initial activity and potential detector saturation need to be managed effectively.

8.3 Data:

Source	Age (sec)	Shield Material	Shield Thickness (mm)	Distance (cm)	Count Duration (sec)	Count
Pa-234	4	none	0	6	3	21771
Pa-234	7	none	0	6	3	21451
Pa-234	11	none	0	6	3	21392
Pa-234	14	none	0	6	3	21257
Pa-234	18	none	0	6	3	20756
Pa-234	21	none	0	6	3	20272
Pa-234	25	none	0	6	3	20178
Pa-234	28	none	0	6	3	19603
Pa-234	32	none	0	6	3	19353
Pa-234	35	none	0	6	3	19193
Pa-234	39	none	0	6	3	18605
Pa-234	42	none	0	6	3	18417
Pa-234	46	none	0	6	3	17798
Pa-234	49	none	0	6	3	17499
Pa-234	53	none	0	6	3	17138
Pa-234	56	none	0	6	3	16803
Pa-234	60	none	0	6	3	16631
Pa-234	63	none	0	6	3	16083
Pa-234	67	none	0	6	3	15648
Pa-234	70	none	0	6	3	15332
Pa-234	74	none	0	6	3	14929
Pa-234	77	none	0	6	3	14656
Pa-234	81	none	0	6	3	14178
Pa-234	84	none	0	6	3	13909
Pa-234	88	none	0	6	3	13709
Pa-234	91	none	0	6	3	13188
Pa-234	95	none	0	6	3	12882
Pa-234	98	none	0	6	3	12469
Pa-234	102	none	0	6	3	11854
Pa-234	105	none	0	6	3	11757
Pa-234	109	none	0	6	3	11376
Pa-234	112	none	0	6	3	11152
Pa-234	116	none	0	6	3	10734

Pa-234	119	none	0	6	3	10287
Pa-234	123	none	0	6	3	9913
Pa-234	126	none	0	6	3	9659
Pa-234	130	none	0	6	3	9285
Pa-234	133	none	0	6	3	9006
Pa-234	137	none	0	6	3	8726
Pa-234	140	none	0	6	3	8293
Pa-234	144	none	0	6	3	8043
Pa-234	147	none	0	6	3	8059
Pa-234	151	none	0	6	3	7461
Pa-234	154	none	0	6	3	7177
Pa-234	158	none	0	6	3	6980
Pa-234	161	none	0	6	3	6889
Pa-234	165	none	0	6	3	6342
Pa-234	168	none	0	6	3	6198
Pa-234	172	none	0	6	3	6142
Pa-234	175	none	0	6	3	5596
Pa-234	179	none	0	6	3	5543
Pa-234	182	none	0	6	3	5482
Pa-234	186	none	0	6	3	5244
Pa-234	189	none	0	6	3	5105
Pa-234	193	none	0	6	3	4746
Pa-234	196	none	0	6	3	4653
Pa-234	200	none	0	6	3	4481
Pa-234	203	none	0	6	3	4127
Pa-234	207	none	0	6	3	3928
Pa-234	210	none	0	6	3	3906
Pa-234	214	none	0	6	3	3740
Pa-234	217	none	0	6	3	3532
Pa-234	221	none	0	6	3	3316
Pa-234	224	none	0	6	3	3292
Pa-234	228	none	0	6	3	3183
Pa-234	231	none	0	6	3	2929
Pa-234	235	none	0	6	3	2853
Pa-234	238	none	0	6	3	2771
Pa-234	242	none	0	6	3	2735
Pa-234	245	none	0	6	3	2647
Pa-234	249	none	0	6	3	2436
Pa-234	252	none	0	6	3	2315
Pa-234	256	none	0	6	3	2293
Pa-234	259	none	0	6	3	2241
Pa-234	263	none	0	6	3	2092
Pa-234	266	none	0	6	3	2015
Pa-234	270	none	0	6	3	1928
Pa-234	273	none	0	6	3	1823
Pa-234	277	none	0	6	3	1754
Pa-234	280	none	0	6	3	1685
Pa-234	284	none	0	6	3	1604
Pa-234	287	none	0	6	3	1640
Pa-234	291	none	0	6	3	1502
Pa-234	295	none	0	6	3	1436
Pa-234	298	none	0	6	3	1419
Pa-234	301	none	0	6	3	1383
Pa-234	305	none	0	6	3	1323
Pa-234	308	none	0	6	3	1196
Pa-234	312	none	0	6	3	1160

Pa-234	316	none	0	6	3	1160
Pa-234	319	none	0	6	3	1046
Pa-234	323	none	0	6	3	1033
Pa-234	326	none	0	6	3	1026
Pa-234	330	none	0	6	3	995
Pa-234	333	none	0	6	3	987
Pa-234	337	none	0	6	3	942
Pa-234	340	none	0	6	3	843
Pa-234	344	none	0	6	3	793
Pa-234	347	none	0	6	3	807
Pa-234	351	none	0	6	3	773



8.4 discussion:

From the plot we can say activity($\boldsymbol{\lambda}$) is $0.01\boldsymbol{s^{-1}}$.
And value of $\boldsymbol{T_{1/2} = 0.693/\lambda = 69.3s}$

9 Half-Lives of Other Sources

In this experiment, you may also encounter other radioactive sources such as Americium-241 (^{241}Am), Strontium-90 (^{90}Sr), and Barium-133 (^{133}Ba). Below are the half-lives of these sources along with explanations regarding their significance.

9.1 Americium-241 (^{241}Am, α Source)

9.1.1 Half-Life

The half-life of Americium-241 is approximately 432.2 years.

9.1.2 Explanation

- **Long Half-Life**: The long half-life of ^{241}Am means that it decays very slowly over time, making it a stable and long-lasting source for experiments involving alpha particles. Due to its slow decay, the activity of the source remains relatively constant over a typical experimental timescale.

- **Safety Considerations**: Despite its long half-life, ^{241}Am emits alpha particles, which can be hazardous if ingested or inhaled. However, alpha particles are easily stopped by a few centimeters of air or a sheet of paper, so proper shielding can effectively mitigate the risk.

9.2 Strontium-90 (^{90}Sr, β Source)

9.2.1 Half-Life

The half-life of Strontium-90 is approximately 28.8 years.

9.2.2 Explanation

- **Moderate Half-Life**: ^{90}Sr has a moderate half-life, meaning it decays at a rate that is fast enough to be measurable over several decades, yet slow enough to remain active for long periods. This makes it useful in experiments that require a sustained β source.
- **Beta Radiation**: ^{90}Sr is a pure beta emitter, which makes it useful for experiments that specifically require beta particles. Its beta particles are more penetrating than alpha particles, requiring more substantial shielding.

9.3 Barium-133 (^{133}Ba, γ Source)

9.3.1 Half-Life

The half-life of Barium-133 is approximately 10.5 years.

9.3.2 Explanation

- **Shorter Half-Life**: ^{133}Ba has a shorter half-life compared to ^{241}Am and ^{90}Sr, which means it decays more rapidly. This makes it more suitable for applications where a gamma source with a relatively short lifespan is desired.
- **Gamma Radiation**: ^{133}Ba emits gamma rays, which are highly penetrating and require significant shielding (such as lead) to protect against. Gamma sources like ^{133}Ba are often used in calibration of gamma detectors and in medical imaging applications.

Conclusion: The half-lives of ^{241}Am, ^{90}Sr, and ^{133}Ba are significantly different, reflecting their varying roles and applications in experiments. ^{241}Am, with its long half-life, provides a stable alpha source, ^{90}Sr offers a moderate-lived beta source, and ^{133}Ba serves as a relatively short-lived gamma source. Understanding the half-life of each source is crucial for selecting the appropriate material for specific experimental needs and for ensuring safe handling and storage.